# E-Commerce Sales Analytics - Executive Overview

Clear all slicers

| $1.31bn | $1.26bn | 3M | $4,209.86 ↑ |
|---|---|---|---|
| Total Revenue | Net Revenue | Units Sold | AOV KPI |

**Year**

- Select all
- 2022
- 2023
- 2024

**Quarter, Month**

All

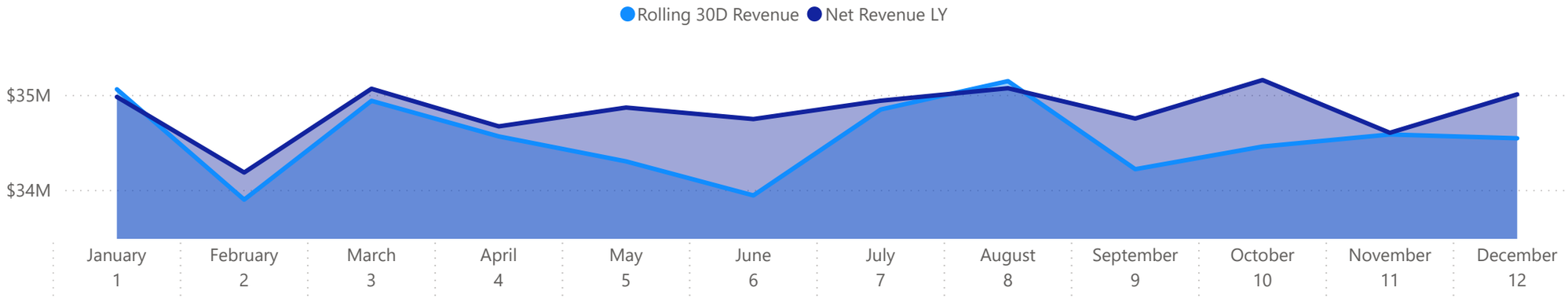
**Category**

All

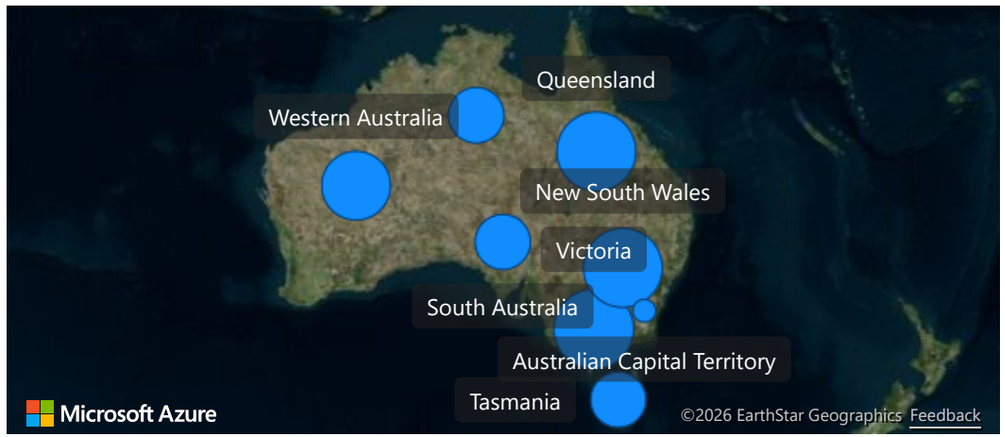
**Region**

All

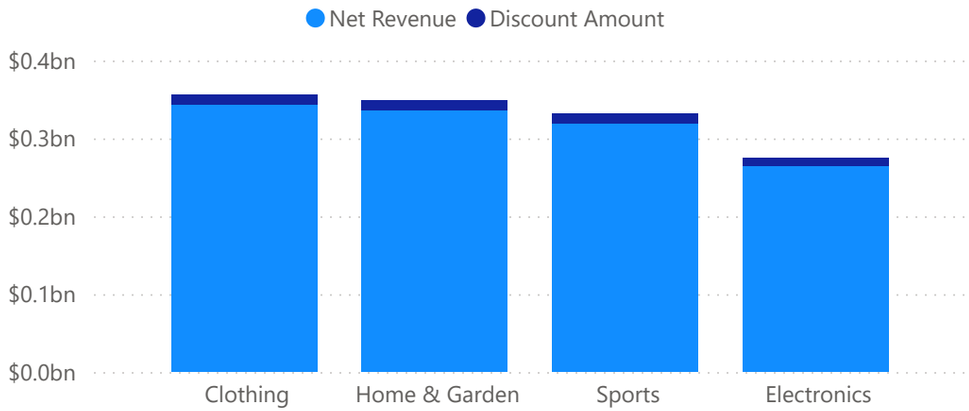
## Net Revenue over Time (Month Hierarchy)




## Revenue by region




## Category contribution



# E-Commerce Sales Analytics - Product & Category Performance

Clear all slicers

| Hill-Hamilton Eco Phone | 100% | 4.49K | $6.57M |
|---|---|---|---|
| Top Product Name | Top Product Contribution % | Top Product - SKU Concentration | Top Product - Revenue |

**Top N**

All

**Brand**

All

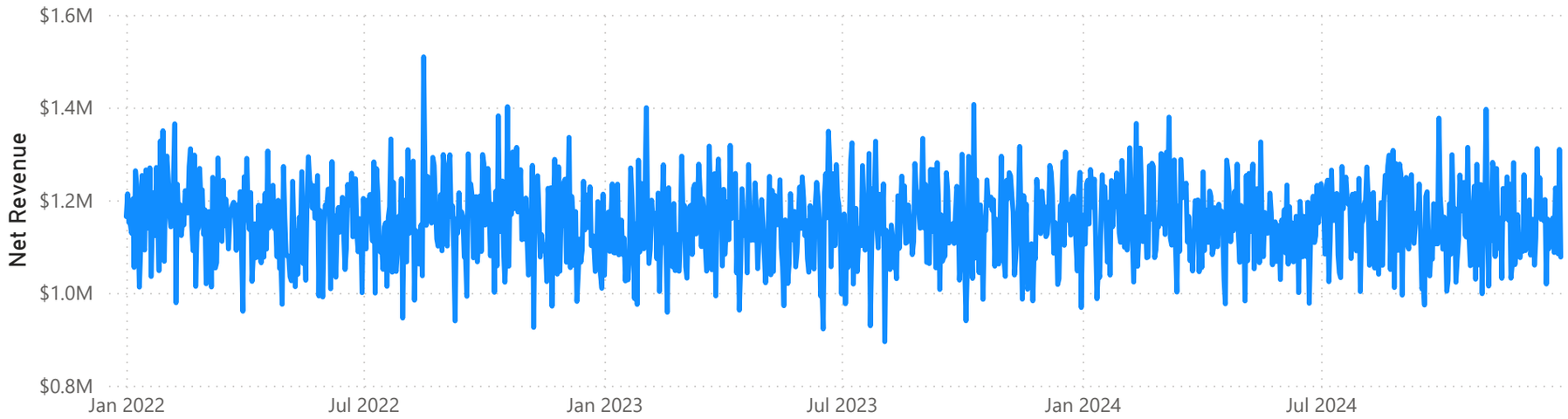
**PromotionName**

All

**ProductName**

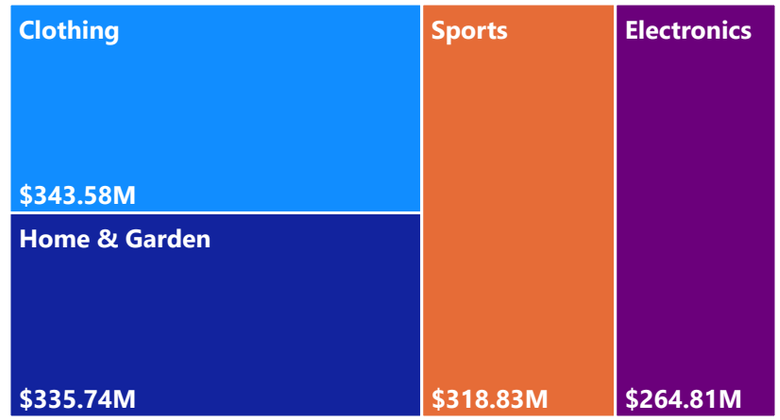
All

## Product trend line




| ProductName | Net Revenue | Units Sold | Discount Amount | Gross Margin | Product Rank |
|---|---|---|---|---|---|
| Hill-Hamilton Eco Phone | $6,289,903.74 | 13498 | $262,343.75 | $2,755,169.23 | 1 |
| Phillips Smart Backpack | $4,671,465.42 | 5365 | $187,069.53 | $2,046,209.97 | 2 |
| Lewis-Hernandez Classic With | $4,602,806.37 | 5384 | $171,910.40 | $2,010,425.76 | 3 |
| Case-Wheeler Tv Ago | $4,529,037.51 | 5097 | $171,227.67 | $1,995,114.93 | 4 |
| Initech Fund And | $4,502,360.93 | 5343 | $185,746.62 | $1,985,914.79 | 5 |
| Stark Classic Shoes | $4,490,019.94 | 8475 | $174,330.76 | $1,963,004.00 | 6 |
| Case-Wheeler Development Ago | $4,419,674.90 | 8178 | $178,423.01 | $1,935,691.85 | 7 |
| Jackson Fund And | $4,367,567.26 | 5563 | $164,558.91 | $1,913,338.55 | 8 |

## Net Revenue by Category and SubCategory



# E-Commerce Sales Analytics - Customer and Geography Analysis

Clear all slicers

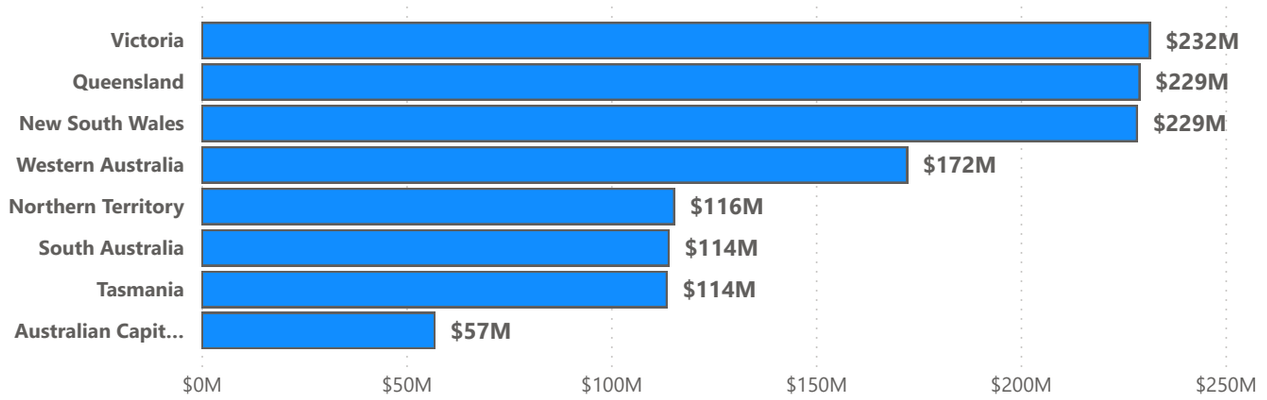
| 31K | 281 | 30K | 99.1% ↑ |
|---|---|---|---|
| Customer Count | New Customers | Returning Customers | Repeat Purchase Rate KPI |

Segment
All

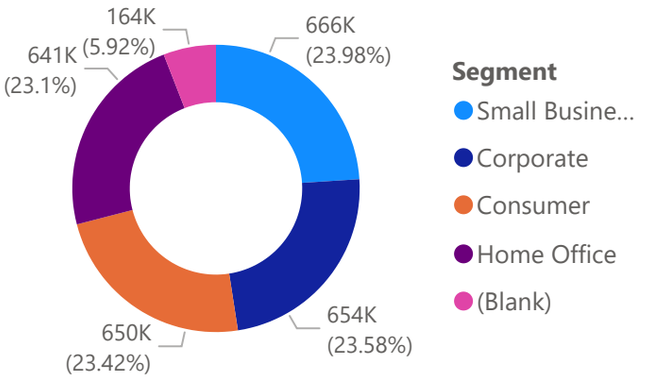
Region
All

State
All

## Net Revenue by State



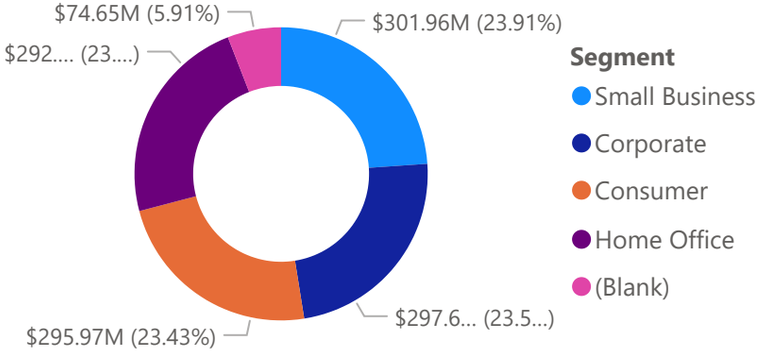

## Units Sold by Segment




## Net Revenue and RPR by City

## New vs. returning trend

## Net Revenue by Segment



# E-Commerce Sales Analytics - Discount & Promotion Impact

Clear all slicers

**PromotionType**

All

**Date**

1/01/2022

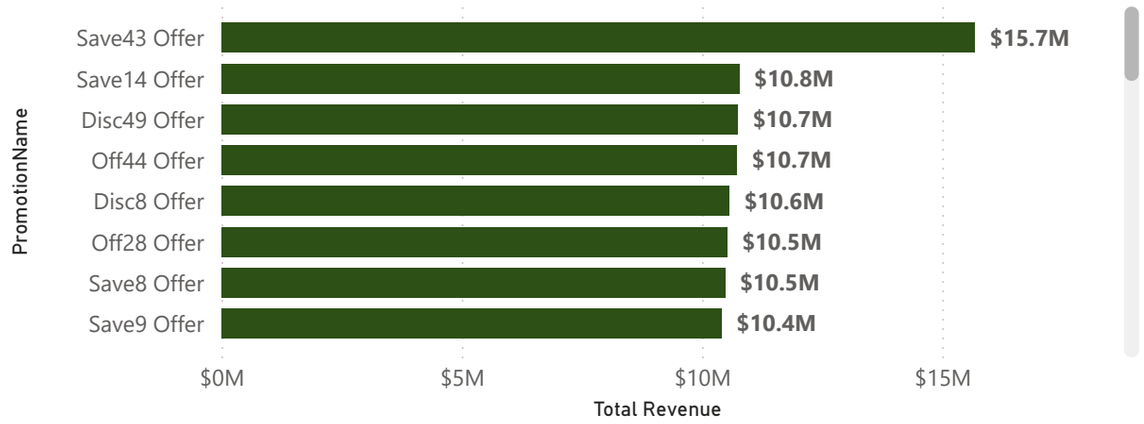
31/12/2024

| $48.37M | 3.68% | $161.22 | $552.55M |
|---|---|---|---|
| Discount Amount | Discount % | Average Discount per Order | Gross Margin |

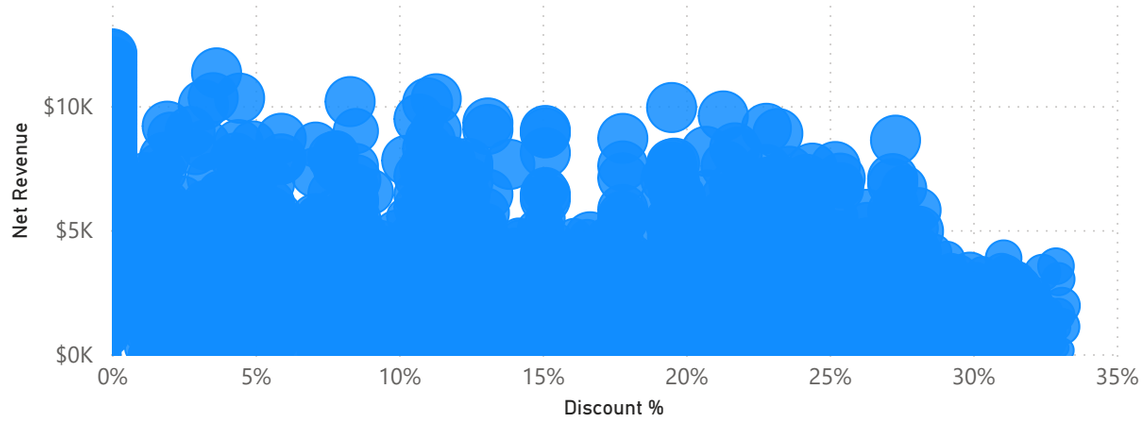
## Promotions Table

| PromotionName | Discount % | Net Revenue | Units Sold | Gross Margin |
|---|---|---|---|---|
| Save43 Offer | 16.32% | $13,074,698.63 | 32865 | $4,617,346.85 |
| Off44 Offer | 15.13% | $9,082,662.29 | 22369 | $3,268,445.50 |
| Deal11 Offer | 12.95% | $8,968,277.88 | 21879 | $3,398,174.70 |
| Save8 Offer | 14.42% | $8,945,430.08 | 22298 | $3,270,073.06 |
| Save9 Offer | 14.45% | $8,879,886.29 | 22004 | $3,297,249.36 |
| Off36 Offer | 15.04% | $8,813,147.88 | 21907 | $3,189,639.46 |
| Off28 Offer | 19.24% | $8,472,883.19 | 21948 | $2,818,957.29 |
| Save14 Offer | 21.68% | $8,416,605.98 | 22429 | $2,629,045.78 |
| Disc8 Offer | 21.27% | $8,290,747.24 | 22514 | $2,563,235.66 |
| Disc22 Offer | 20.76% | $8,000,553.45 | 21796 | $2,526,650.69 |
| Off45 Offer | 21.61% | $7,994,834.64 | 21939 | $2,448,234.69 |
| Disc49 Offer | 26.53% | $7,865,422.38 | 22507 | $2,083,125.08 |
| Off37 Offer | 7.23% | $4,962,825.12 | 11220 | $2,064,538.51 |
| Off26 Offer | 8.65% | $4,921,898.09 | 11238 | $1,983,368.59 |
| Deal26 Offer | 8.41% | $4,827,554.85 | 11190 | $1,967,190.02 |
| Disc6 Offer | 6.06% | $4,806,087.57 | 10961 | $2,024,806.18 |
| Deal24 Offer | 8.66% | $4,793,499.88 | 11029 | $1,942,379.03 |
| Promo34 Offer | 11.17% | $4,788,813.23 | 11402 | $1,871,587.13 |
| Deal25 Offer | 12.57% | $4,681,484.24 | 11326 | $1,775,131.83 |
| Disc21 Offer | 11.46% | $4,589,430.63 | 11041 | $1,782,054.00 |
| Save23 Offer | 11.94% | $4,526,111.35 | 10860 | $1,739,137.58 |
| Save15 Offer | 13.25% | $4,471,085.02 | 10862 | $1,695,615.59 |
| Deal29 Offer | 19.75% | $4,436,492.59 | 11340 | $1,464,334.43 |

## Net Revenue vs Discounts by Promotion




## Revenue, Discounts & Units by Order Line



# E-Commerce Sales Analytics - Data Quality & Validation

Clear all slicers

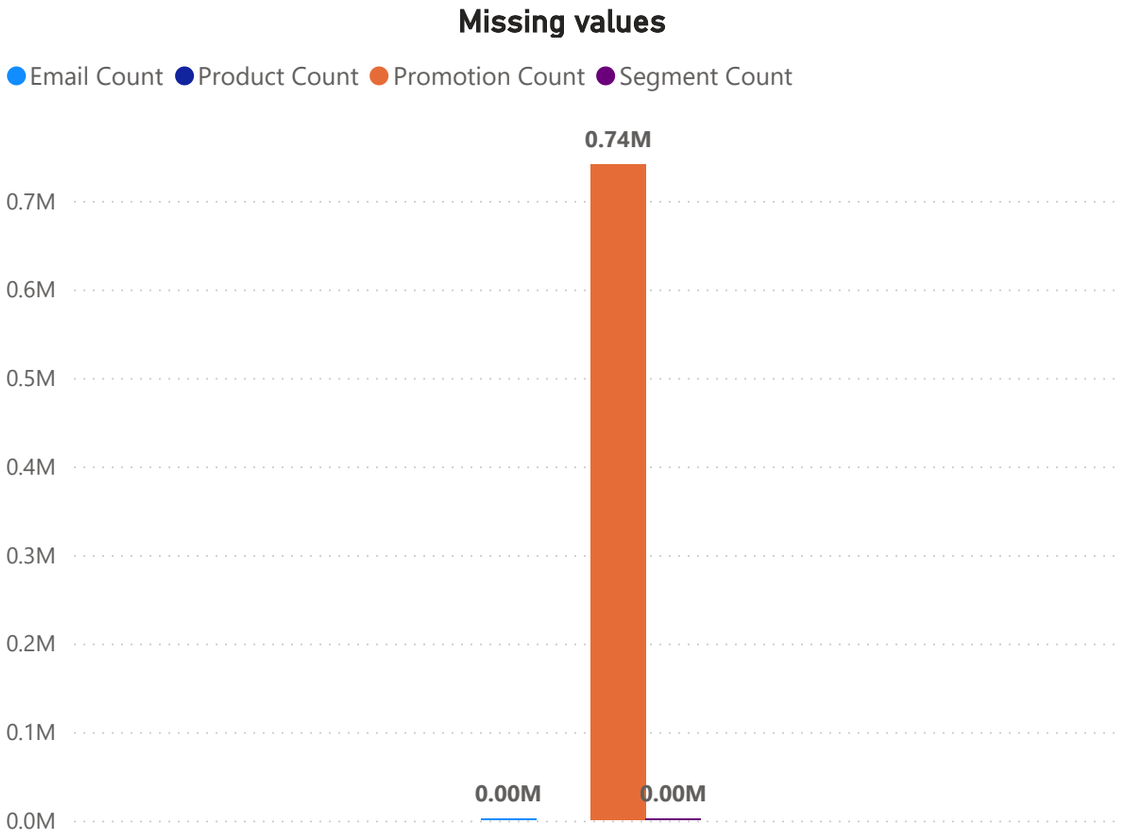
| 925332 | 1096 | 23 | 30600 | 50 | 300000 |
|---|---|---|---|---|---|
| Total Sales | Total Days Recorded | Total Categories | Total Customers | Total Promotions | Total Orders |

| 22 | 1030 | 2024-12-31 |
|---|---|---|
| Total Geographies | Total Products | Last Refreshed |

| Customers | Days | Geographies | Orders | Products | Promotions |
|---|---|---|---|---|---|
| 30596 | 1096 | 22 | 300000 | 1030 | 51 |

## ETL & Data Quality Summary

Data was ingested into staging tables (stg_*) and transformed into clean reporting tables (clean_*) using Power Query. Key cleaning steps included removing duplicates, trimming text, applying Proper Case to names and categories, standardizing data types (currency, percentage, date), and handling nulls. Enrichments included deriving product brands, standardizing promotion types, and filling missing customer emails with "No Email." Staging and raw tables were disabled post-validation to optimize the model.
Calculated fields and measures were created in DAX to support margin analysis, customer growth, category revenue, AOV, and trend indicators. A margin proxy was computed where product cost data existed, with remaining calculations handled in DAX.

**Data Quality Limitations:**

- ~80% of PromotionKeys in sales data are null, limiting promotion-level analysis
- Incomplete product cost data affects margin accuracy
- ~4% of Segment values in DimCustomer are null
- Some customer email data is synthetic
- Minor join mismatches exist: 30,600 CustomerKeys in clean_DimCustomer vs 30,596 in clean_FactOrders (4 customers had no orders), and 50 PromotionKeys in clean_DimPromotions vs 51 mapped in clean_FactSales (historical or inactive promotions in sales data)